Directory of proofs

This directory indicates where to find the proofs from the lessons. It does not fix a universal number of “examinable proofs”: the groupings vary according to the presentation.

Result and approach	File
Induction: explicit form, sum, divisibility	Ch. 1, p. 1
Increasing sequence not bounded above; comparison towards infinity	Ch. 2, p. 15
Bernoulli and limits of q^n	Ch. 2, p. 15
Limits of the exponential and comparison of growth rates	Ch. 3, p. 22
Convex curve above its tangents	Ch. 4, p. 36
Derivative of ln; limit of $x \ln x$ at 0^+	Ch. 6, p. 51
Two antiderivatives differ by a constant; $y' = ay$	Ch. 7, p. 65
Antiderivative of an area (increasing case); integration by parts	Ch. 8, p. 72
Pascal by two methods; number of subsets	Ch. 9, p. 29
Orthogonal projection and minimum distance	Ch. 11, p. 58
Cartesian equation from a normal vector	Ch. 12, p. 87
Expression of the binomial distribution	Ch. 13, p. 44
Binomial expected value and variance; concentration and LLN	Ch. 14, p. 80